

CS 544 (Summer 2026): Project 3 Report

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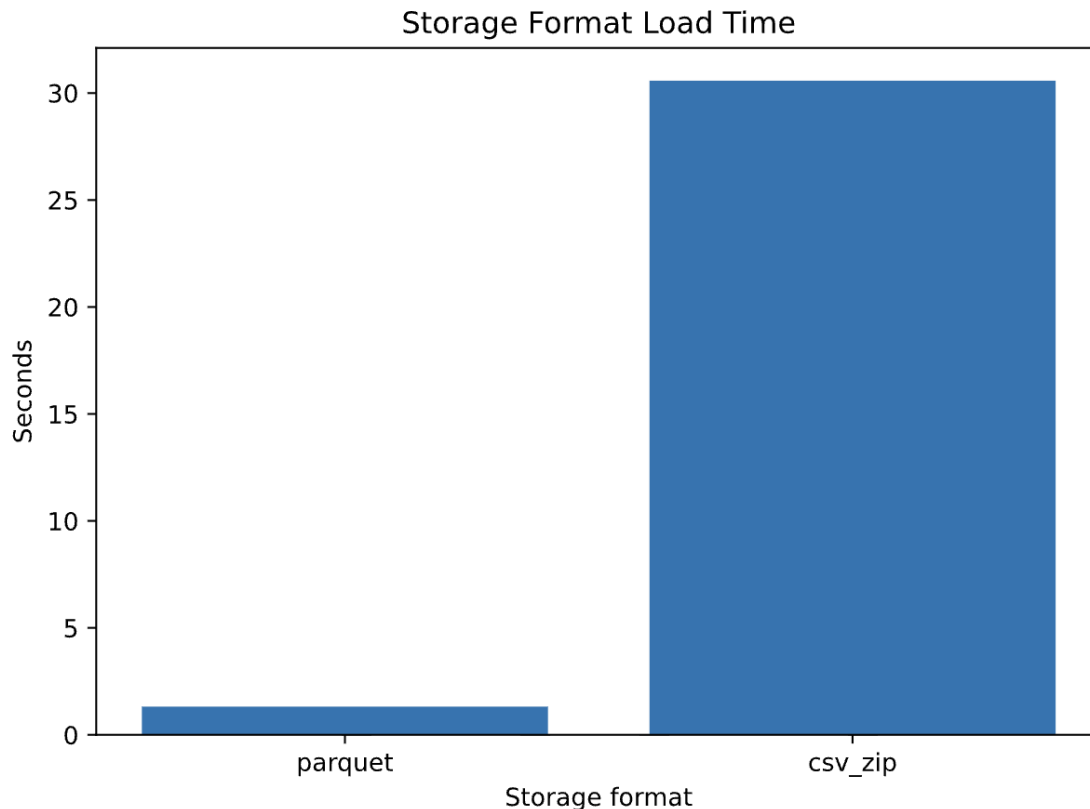
Don't change the number of pages, and don't let content for a specific part exceed the assigned page.

Part 2: Storage Performance

In your own words, what question are we asking with this experiment?

Which storage format loads data faster for this application: Parquet or zipped CSV?

Plot here:



How do you interpret the results? Use the plot to answer the question you posed above. Be sure to reference at least one actual number in your specific plot as part of your answer.

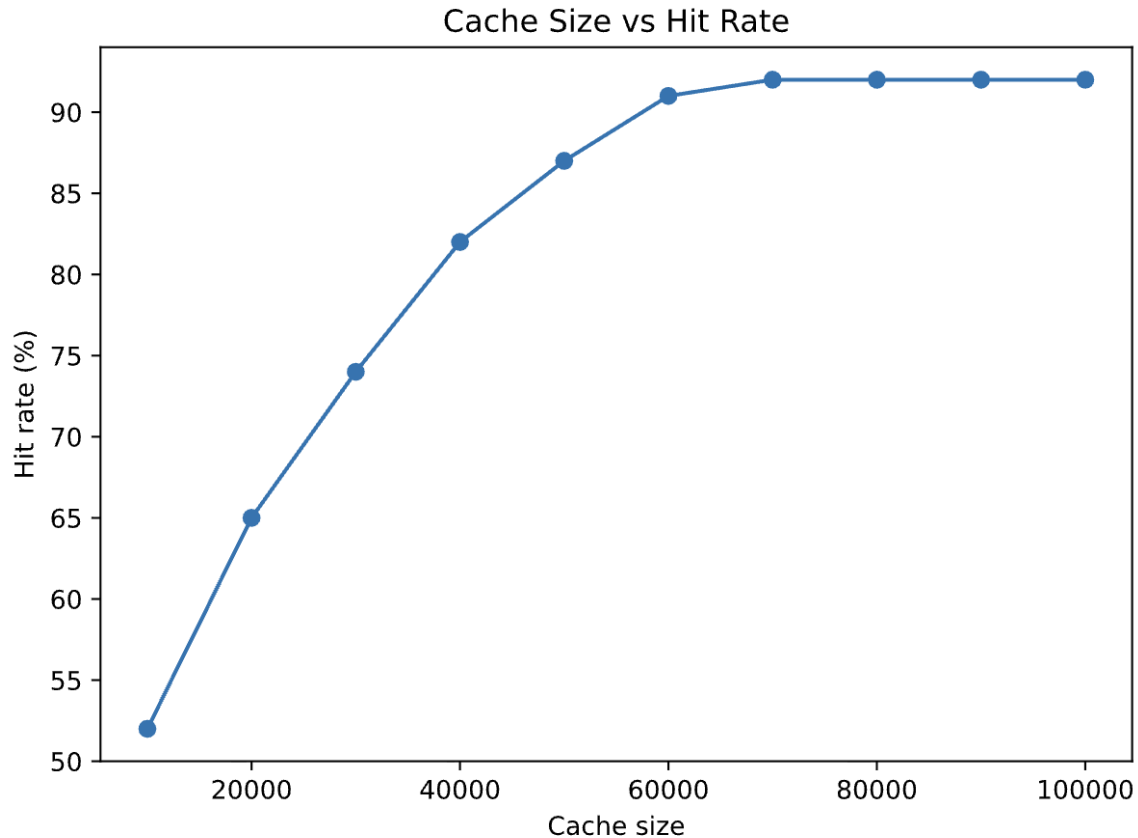
The results show that Parquet is much faster than the zipped CSV format. In my experiment, Parquet loaded the data in about 1.32 seconds, while the zipped CSV took about 30.58 seconds. This indicates that Parquet is much more efficient for loading the selected columns because it is a column-oriented binary format, while the CSV file must be decompressed and parsed as text.

Part 3: Memory Performance

In your own words, what question are we asking with this experiment?

How does cache size affect the HTTP cache hit rate during the analysis?

Plot here:



How do you interpret the results? Use the plot to answer the question you posed above. Be sure to reference at least one actual number in your specific plot as part of your answer.

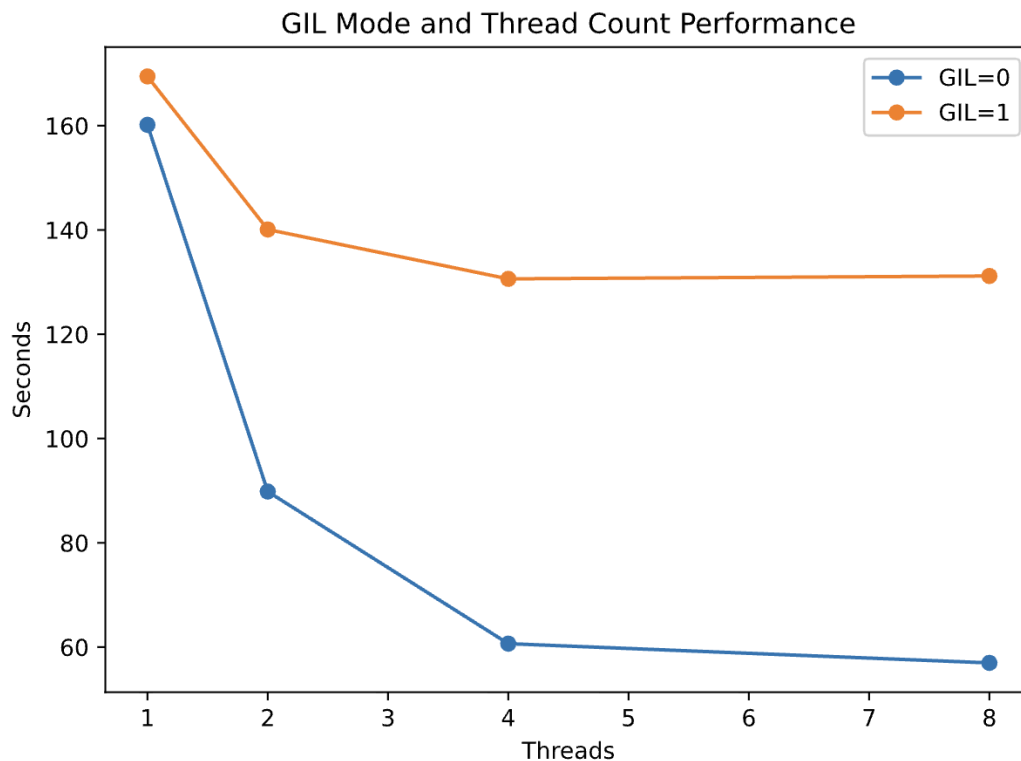
The plot shows that increasing the cache size improves the cache hit rate. The hit rate increases from 52% with a cache size of 10,000 to 92% with a cache size of 70,000. After that, increasing the cache size further provides little improvement because most frequently accessed census tracts are already stored in the cache.

Part 4: Compute Performance

In your own words, what question are we asking with this experiment?

How do the number of threads and the Python GIL affect the running time of the client program?

Plot here:



How do you interpret the results? Use the plot to answer the question you posed above. Be sure to reference at least one actual number in your specific plot as part of your answer.

With the GIL enabled, increasing the number of threads only provides a small performance improvement. The runtime decreases from about 169.44 seconds with one thread to about 131.17 seconds with eight threads. With the GIL disabled, the program scales much better. The runtime decreases from about 160.16 seconds with one thread to about 56.97 seconds with eight threads, showing that true parallel execution significantly improves performance.